Project Documentation Package

🎛️ Part 1: User Guide (One-Page Layout)

🔄 1. Data Refresh Instructions

- ✓ Manual Desktop Refresh: Open the .pbix file in Power BI Desktop. Go to the Home tab on the top ribbon and click the Refresh button to reload Hospital_General_Information.csv.
- ✓ Scheduled Cloud Refresh: Once published to the Power BI Service workspace, navigate to the dataset settings, expand Scheduled Refresh, switch the toggle to On, and configure a daily schedule (e.g., daily at 8:00 AM) to automatically keep records updated.

🎛️ 2. Report Navigation & Slicers

- ✓ Page Layout: The report is structured into five distinct sections: Overview, Geographic Analysis, Performance Measure Details, Detailed Hospital View, and Scenario Insights.
- ✓ Global Slicers: Use the dropdown slicers at the top of the report pages to filter records dynamically by State, Hospital Type, and Hospital Ownership.
- ✓ Clearing Filters: Click the small Clear Selections (eraser) icon on the top-right corner of any slicer to reset the filters and return to a comprehensive nationwide data view.

📈 3. Key Performance Indicators (KPIs)

- ✓ Total Hospitals Count: Displays the gross count of all active healthcare facilities currently loaded in the model.
- ✓ Avg% Bettr (Safety and ReadM): Represents the cross-domain composite baseline tracking the combined quality scores for safety and readmissions.
- ✓ % Birthing-Friendly: Tracks the exact proportion of hospitals nationwide meeting specialized criteria for birthing-friendly designations.

🎚️ 4. "What-If" Scenario Sliders (Scenario Insights Page)

- ✓ Rating Threshold Slicer: Move this slider to define what qualifies as a high-performing facility. Setting it to 4 dynamically classifies ratings of 4 and 5 as High-Performing, and 1–3 as Low-Performing.
- ✓ Measure Improvement Slicer: Adjust this percentage slider (from 0% to 100%) to simulate a specific percentage shift of underperforming hospitals out of the "Worse" quality tier.
- ✓ Interactive Outputs: The Projected High-Performing Hospitals metric card and trend lines on the Scenario Insights page will recalculate dynamically as you move the slider.

Part 2: Technical Document

1. Entity Relationship Diagram (ERD) Structure

The data model uses an optimized Star Schema centered around a core fact table and connected lookup dimensions:

• Fact Table: Hospital (Primary Key: Facility ID)

• Dimension Tables (1:N Relationships via Direct Keys):

- • State_Lookup connected via the State column.
- • County_Lookup connected via the County/Parish column.
- • HospitalType_Lookup connected via the Hospital Type column.
- • Ownership_Lookup connected via the Hospital Ownership column.
- • PerformanceDomain_Bridge (Unpivoted table for domain-level measure distributions).

2. Power Query Data Transformation Steps

a. Source Ingestion: Connected directly to Hospital_General_Information.csv with source text encoding verified.
b. Column Standardization: Cleaned and sanitized all 38 original column headers, removing spaces, slashes, and special characters to ensure robust syntax.
c. Data Cleaning & Type Casting:
   - Replaced non-numeric placeholder values in Hospital overall rating with Null blanks and cast the column as a whole number integer.
   - Standardized the EmergencyServices field into a uniform, logical Yes/No binary string layout.
   - Applied Text.Trim globally across all text-based fields to remove erroneous leading or trailing white spaces.

3. Core DAX Formulas

✓ Total Hospitals Count:

dax

TotalHospitalsCount = COUNTROWS('Hospital')

✓ Average % Better (Safety & Readmissions KPI):

dax

Avg% Bettr (Safety and ReadM) = DIVIDE ([Safety % Better] + [READM % Better], 2, 0)

✓ Projected High-Performing Hospitals:

dax

Projected High-Performing Hospitals =

```
CALCULATE ([TotalHospitalsCount], FILTER('Hospital', 'Hospital'[Hospital overall rating] >= [RatingThresholdValue]) ) + ([LowPerformingHospitalsCount] * [MeasureImprovementValue])

MEASURE _Measures[Shift Share to Better] = 0.5

MEASURE _Measures[High-Performing Hospitals] =

  VAR T = [Rating Threshold Value]

  RETURN CALCULATE([Total Hospitals], Hospital[Hospital overall rating] >= T)

MEASURE _Measures[Low-Performing Hospitals] = [Rated Hospitals] - [High-Performing Hospitals]

MEASURE _Measures[% High-Performing] = DIVIDE([High-Performing Hospitals], [Rated Hospitals])

MEASURE _Measures[% Low-Performing] = DIVIDE([Low-Performing Hospitals], [Rated Hospitals])

MEASURE _Measures[Projected High-Performing Hospitals] =

  [High-Performing Hospitals] + [Measure Improvement Value] * [Low-Performing Hospitals]

MEASURE _Measures[Projected % High-Performing] = DIVIDE([Projected High-Performing Hospitals], [Rated Hospitals])


MEASURE _Measures[MORT Projected Worse] = [MORT Worse] * (1 - [Measure Improvement Value])

MEASURE _Measures[MORT Projected Better] = [MORT Better] + [MORT Worse] * [Measure Improvement Value] * [Shift Share to Better]

MEASURE _Measures[MORT Projected No Different] = [MORT No Different] + [MORT Worse] * [Measure Improvement Value] * (1 - [Shift Share to Better])

MEASURE _Measures[MORT Projected % Better] = DIVIDE([MORT Projected Better], [MORT Measures Reported])

MEASURE _Measures[MORT Projected % No Different] = DIVIDE([MORT Projected No Different], [MORT Measures Reported])

MEASURE _Measures[MORT Projected % Worse] = DIVIDE([MORT Projected Worse], [MORT Measures Reported])


MEASURE _Measures[Safety Projected Worse] = [Safety Worse] * (1 - [Measure Improvement Value])

MEASURE _Measures[Safety Projected Better] = [Safety Better] + [Safety Worse] * [Measure Improvement Value] * [Shift Share to Better]

MEASURE _Measures[Safety Projected No Different] = [Safety No Different] + [Safety Worse] * [Measure Improvement Value] * (1 - [Shift Share to Better])

MEASURE _Measures[Safety Projected % Better] = DIVIDE([Safety Projected Better], [Safety Measures Reported])
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MEASURE _Measures[Safety Projected % No Different] = DIVIDE([Safety Projected No Different], [Safety Measures Reported])

MEASURE _Measures[Safety Projected % Worse] = DIVIDE([Safety Projected Worse], [Safety Measures Reported])

MEASURE _Measures[READM Projected Worse] = [READM Worse] * (1 - [Measure Improvement Value])

MEASURE _Measures[READM Projected Better] = [READM Better] + [READM Worse] * [Measure Improvement Value] * [Shift Share to Better]

MEASURE _Measures[READM Projected No Different] = [READM No Different] + [READM Worse] * [Measure Improvement Value] * (1 - [Shift Share to Better])

MEASURE _Measures[READM Projected % Better] = DIVIDE([READM Projected Better], [READM Measures Reported])

MEASURE _Measures[READM Projected % No Different] = DIVIDE([READM Projected No Different], [READM Measures Reported])

MEASURE _Measures[READM Projected % Worse] = DIVIDE([READM Projected Worse], [READM Measures Reported])

MEASURE _Measures[Avg % Better (Safety & READM)] =

AVERAGEX({ [Safety Projected % Better], [READM Projected % Better] }, [Value])

4. Instructions for Adding New Performance Domains

To scale the architecture for additional quality metrics (e.g., Timeliness of Care - TE):

I. Open the Power Query Editor, navigate to the raw source data table, and identify the measure count columns for the target domain.
II. Unpivot the new domain's "Better", "No Different", and "Worse" count fields into the existing structured PerformanceDomain_Bridge table layout.
III. Click apply, then write three corresponding base measures in DAX to calculate percentages: PctTEBetter, PctTENoDifferent, and PctTEWorse.
IV. Incorporate the new domain series directly into the clustered column charts found on the Performance Measure Details and Detailed Hospital View pages.